

Aakash Vanmali



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🌐 [linkedin.com/in/aakashvanmali](https://www.linkedin.com/in/aakashvanmali) 🏠 github.com/aakashvanmali45 🌐 Portfolio

Education

TH Deggendorf, Germany, M.Eng. in Mechatronics and Cyber-Physical Systems Mar 2024 – Present

- Current Grade: 2.0
- **Key Coursework:** Autonomous Systems, Advanced Robotics, Modelling & Simulation, Additive Manufacturing, VR/AR Interfaces
- **Thesis:** Automated Analysis of Technical Diagrams Using Hybrid Machine Learning Models

University of Mumbai, India, B.Eng. in Mechanical Engineering Jun 2019 – Jun 2023

- Final Grade: 1.6
- **Bachelor Thesis:** Deep Learning and Computer Vision for Self-Driving Cars
- **Coursework:** CAD/CAM, Industrial Electronics, Automation & AI, Python Programming

Experience

Master Thesis, ABB AG. - Mannheim, Germany 01/2026 – current

- Researching unsupervised clustering and classification methods for automated analysis of technical diagrams, focusing on scalable approaches that generalize without labeled data.
- Investigating hybrid machine learning models combining feature extraction, dimensionality reduction, and probabilistic clustering for structured document understanding.

Computer Vision and Machine Learning Intern, ABB AG. - Mannheim, Germany 09/2025 – 12/2025

- Designed an unsupervised pipeline for estimating optimal cluster count from learned feature embeddings without prior knowledge. Validated on MNIST, correctly recovering 10 classes (ARI = 0.76, NMI = 0.84).
- Explored open-world and open-vocabulary symbol detection using an OWL-based detector to enable zero-shot recognition of unseen symbol classes, reducing dependency on annotated training data.
- Developed a symbol-to-text spatial association method for linking detected P&ID symbols to their nearest relevant text labels, enabling structured information extraction from engineering drawings.
- Investigated image pre-processing strategies (adaptive rescaling) to improve OCR and object detection robustness on degraded, low-resolution scanned industrial documents.
- Integrated Azure Document Intelligence into the processing pipeline, benchmarking it against open-source alternatives (EasyOCR) for production-grade text detection on engineering drawings.
- **Skills:** Python, PyTorch, ResNet, UMAP, GMM, EasyOCR, Azure Document Intelligence, OWL-ViT, OpenCV

Computer Vision Intern, JSW Steel Ltd. - Mumbai, India 08/2023 – 12/2023

- Employed video analytics techniques to evaluate steel industry-specific use cases.
- Developed and implemented a project aimed at accurately measuring gas consumption in the steel industry using non-contact methods, eliminating inaccuracies caused by flowmeter choking.
- Utilized Computer Vision and Convolutional Neural Networks to analyze images and determine the volume of flame flared from flare stacks.
- Leveraged Docker for packaging the solution into an end-to-end application, ensuring seamless deployment and integration.
- **Skills:** Computer Vision, Convolutional Neural Network, Docker, OpenCV, Tensorflow, Pandas

Technical Projects

Chain Wheeled Mobile Robot with IR Camera for Object Identification

[Project Link](#)

- Designed a 4-wheel skid-steered mobile robot in Fusion 360 and converted the model to URDF for ROS Noetic integration.
- Integrated Gazebo plugins for LIDAR and camera sensors, and implemented skid-steering kinematics.
- Developed Python ROS nodes for autonomous navigation using LIDAR data and object detection using the SIFT algorithm.
- Achieved reliable autonomous driving and 100% object recognition accuracy in simulation.

Color Sorting Robotic Arm using MATLAB

[Project Link](#)

- Designed a 4R robotic arm in SolidWorks and simulated its motion using MATLAB Simulink and Simscape, applying forward and inverse kinematics with DH parameters.
- Implemented image processing to identify and locate red, green, and blue objects, converting pixel data into real-world coordinates.
- Developed a path planner and vacuum gripper control logic to autonomously perform pick-and-place.
- Achieved reliable simulation performance with accurate color-based sorting, demonstrating effective integration of robotics, image processing, and control systems.

Languages Proficiency

German B1
English C1

Technical Skills

Programming Languages: Python, MATLAB

Deep Learning: PyTorch, TensorFlow, ResNet, CNN architectures, unsupervised/self-supervised learning

Computer Vision: OpenCV, Open3D, object detection (OWL-ViT), OCR (EasyOCR, Azure Doc Intelligence)

Robotics: ROS2 Jazzy, ROS Noetic, Gazebo

Training and Certifications

- **ROS2 for Beginners** – Udemy, 2025
- **Machine Learning for All** – Coursera, 2021
- **Python for Everybody Specialization** – Coursera, 2021

Extracurricular Activities

- **DILE Member** – University social outreach initiative, 2023–2024
- **AI Head**, Google Developer Student Club – Led AI sessions and projects, 2021–2022